Factorising - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Factorise $5x + 15$

Common factor 5

Answer: 5(x + 3)

Question 2 *Practice question written for this worksheet (not from a past paper)*

Factorise $x^2 - 49$

Difference of two squares: $x^2 - 7^2$

Answer: (x - 7)(x + 7)

Question 3 *Practice question written for this worksheet (not from a past paper)*

Factorise $6x^2 + 4x$

Common factor $2x$

Answer: 2x(3x + 2)

Question 4 *Practice question written for this worksheet (not from a past paper)*

Factorise completely $12ab - 8b$

Common factor $4b$

Answer: 4b(3a - 2)

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Factorise $4y^2 - 25$

$(2y)^2 - 5^2$

Answer: (2y - 5)(2y + 5)

Question 6 *Practice question written for this worksheet (not from a past paper)*

Factorise $16x^2 - 9y^2$

$(4x)^2 - (3y)^2$

Answer: (4x - 3y)(4x + 3y)

Question 7 *Practice question written for this worksheet (not from a past paper)*

Factorise $x^2 + 7x + 12$

Numbers with product 12 and sum 7: 3 and 4

Answer: (x + 3)(x + 4)

Question 8 *Practice question written for this worksheet (not from a past paper)*

Factorise $y^2 + 6y$

Common factor y

Answer: y(y + 6)

Question 9 *Practice question written for this worksheet (not from a past paper)*

Factorise $x^2 - 3x - 10$

Product -10, sum -3: -5 and 2

Answer: (x - 5)(x + 2)

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Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2019 · Question 5

5			$(x+4)(x-9)$	2	M1 A1	For $(x+a)(x+b)$ where $ab = -36$ and a and b are integers Ignore extension to roots $x = -4$ & 9
						Total 2 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 8

Q	Working	Answer	Mark	Notes
			2	M1 for $(x \pm 4)(x \pm 8)$ **or** for $(x+a)(x+b)$ with $ab = 32$ or $a+b=-12$
	Correct answer scores full marks (unless from obvious incorrect working)	$(x-4)(x-8)$		A1 for correct factors oe eg $(-x+8)(-x+4)$
				Total 2 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 13

13.	$(x-5)\{4(x-5)+3\}$	$(x-5)(4x-17)$	2	M1 A1	Accept $(x-5)\{4x-20+3\}$ or reaching $4x^2-37x+85$
					Total 2 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 19

19		$8(4x-y)(4x+y)$	2	B2	B2 for $8(4x-y)(4x+y)$ oe B1 for correct, incomplete factorisation eg $(16x-4y)(8x+2y)$ **or** eg $8(16x^2-y^2)$ **or** correct use of difference of two squares eg. $(12x-y-(4x-3y))(12x-y+4x-3y)$
					Total 2 marks

Question 14

Practice question written for this worksheet (not from a past paper)

Factorise $3x^2 + 14x + 8$

$a \times c = 24$; numbers 12 and 2

$3x^2 + 12x + 2x + 8 = 3x(x+4) + 2(x+4)$

Answer: (3x + 2)(x + 4)

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Question 15 *Practice question written for this worksheet (not from a past paper)*

Factorise $2x^2 - 5x - 12$

$a \times c = -24$; numbers -8 and 3

$2x^2 - 8x + 3x - 12 = 2x(x - 4) + 3(x - 4)$

Answer: (2x + 3)(x - 4)

Question 16 *Practice question written for this worksheet (not from a past paper)*

Factorise $5x^2 - x - 4$

$a \times c = -20$; numbers -5 and 4

$5x^2 - 5x + 4x - 4 = 5x(x - 1) + 4(x - 1)$

Answer: (5x + 4)(x - 1)

Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 14*

Q	Working	Answer	Mark	Notes
		$2(5g + 3)(5g - 3)$	3	B3 for $2(5g + 3)(5g - 3)$ B2 for $2(5g \pm 3)(5g \pm 3)$ oe eg $2(5g - 3)^2$ B1 for $2(25g^2 - 9)$ or $(10g + 6)(5g - 3)$ or $(5g + 3)(10g - 6)$ or $(5g + 3)(5g - 3)$
				Total 3 marks

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CHALLENGE QUESTIONS

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 14*

14	(a)		$(a-b)(a+b)$	1	B1 oe
	(b)	$(2^{11}-1)(2^{11}+1)$ **or** $(2048-1)(2048+1)$ **or** $\sqrt{4194304}=2048$ **or** $\sqrt{2^{22}}=2048$ **or** $\sqrt{2^{22}}=2^{11}$ **or** $\sqrt{4194304}=2^{11}$ **or** 3 , 23 , 89 , 683 (may be seen in a factor tree)		2	M1
			2047, 2049		A1 cao
					Total 3 marks

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 12*

Q	Working	Answer	Mark	Notes
(a)	$(2y \pm 1)(3y \pm 4)$ or $(2y \pm 4)(3y \pm 1)$ or $2y(3y-4)+1(3y-4)$ or $3y(2y+1)-4(2y+1)$		2	M1 NB factors must be in the form $(ay+b)$ where a and b are integers Condone use of a different letter to y
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$(2y+1)(3y-4)$		A1 or $(3y-4)(2y+1)$ Ignore further working if solving a quadratic to find roots.
(b)	$\frac{3(2x+1)}{12x}+\frac{4(7-5x)}{12x}$ **or** $\frac{3x(2x+1)}{12x^2}+\frac{4x(7-5x)}{12x^2}$ **or** $\frac{3(2x+1)+4(7-5x)}{12x}$ oe **or** $\frac{3x(2x+1)+4x(7-5x)}{12x^2}$ oe		3	M1 for two correct fractions with common denominator with the intention to add **or** a single correct fraction NB $12x$ can be written as $(3)(4x)$ or $(4)(3x)$ for this mark or $12x^2$ can be written as $(3x)(4x)$ for this mark
	$\frac{6x+3+28-20x}{12x}$ oe or $\frac{6x^2+3x+28x-20x^2}{12x^2}$ oe or $\frac{31x-14x^2}{12x^2}$ oe			M1 for a correct single fraction with **all** brackets expanded
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$\frac{31-14x}{12x}$		A1 or $\frac{14x-31}{-12x}$
				Total 5 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 9

9	(a)		7	1	B1	
	(b)		$8m^{15}p^{12}$	2	B2	for $8m^{15}p^{12}$ (B1 for a single product with any two of 8, m^{15} and p^{12} correct, eg $km^{15}p^{12}$ where $k \neq 8$ **or** $8m^{a}p^{12}$ where $a \neq 15$ **or** $8m^{15}p^{b}$ where $b \neq 12$. One term can be missing with 2 correct for B1)
	(c)		$6cw^2(3c^4 + 5w)$	2	B2	for the correct factorisation (B1 for a correct partial factorisation of at least 2 from 6, c and w or w^2 outside the bracket, eg $6c(3c^4w^2 + 5w^3)$ or $cw(18c^4w + 30w^2)$ **or** for the correct highest common factor on the outside and two terms inside the bracket with one correct eg $6cw^2(3c^4 + \ldots)$ or $6cw^2(\ldots + 5w)$ **or** for $(3c^4 + 5w)$ as a factor)
	(d)	$(x \pm 9)(x \pm 3)$ or $x(x-3) + 9(x-3)$ or $x(x+9) - 3(x+9)$ **OR** $(x+a)(x+b)$ where $ab = -27$ or $a + b = 6$ and a and b are integers and $a, b \neq 0$		2	M1	allow any letter for x
		Correct answer scores full marks (unless from obvious incorrect working)	$(x+9)(x-3)$		A1	correct factorisation; allow any letter for x ISW if correct factorisation is seen followed by an attempt to solve
						Total 7 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 7

Q	Working	Answer	Mark	Notes
(a)	$20 - 5x\ (= 7 - 3x)$		3	M1 for expansion of bracket
	E.g. $20 - 7 = -3x + 5x$ or $-5x + 3x = 7 - 20$			M1 ft from a 4-term equation for a correct process of isolating terms in x on one side of the equation and numbers on the other side
		6.5 oe		A1 dep on M1 awarded and from correct working
(b)			2	M1 for any **correct** partial factorisation with at least 2 factors, one of which must be a letter **or** the correct common factor with no more than 1 error inside the bracket
		$8m^2g^3(2m + 3g^2)$		A1
(c)(i)	$(y \pm 6)(y \pm 8)$		2	M1
		$(y-8)(y+6)$		A1
(c)(ii)		8, −6	1	B1 **must ft from their factors in (c)(i)**
				Total 8 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 17

Q	Working	Answer	Mark	Notes
(a)	$6y(y-1)+5(y-1)$ or $y(6y+5)-1(6y+5)$		2	M1 for $(6y\pm5)(y\pm1)$ or $(6y\pm1)(y\pm5)$ or $(ay+5)(by-1)$ where $ab=6$ or $5b-a=-1$ or $(6y+p)(y+q)$ where $pq=-5$ or $6q+p=-1$ Condone use of a different letter to y
	Correct answer scores full marks (unless from obvious incorrect working)	$(6y+5)(y-1)$		A1 oe
(b)	$8w-fw=2f+3$ oe		3	M1 for multiplying by denominator and expanding in a correct equation
	$8w-3=2f+fw$ oe or $-2f-fw=3-8w$ oe			M1 for gathering terms in f on one side and other terms the other side in a correct equation ft their equation dep on 2 terms in f and two other terms
	Correct answer scores full marks (unless from obvious incorrect working)	$f=\frac{8w-3}{2+w}$		A1 oe accept $f=\frac{3-8w}{-2-w}$ oe
(c)	$4(x^2-2x)+7$ or $4\left(x^2-2x+\frac{7}{4}\right)$ oe		3	M1
	$4\left[(x-1)^2-1^2\right]+7$ oe or $4\left[(x-1)^2-1^2+\frac{7}{4}\right]$ oe			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)	$4(x-1)^2+3$		A1 allow $a=4$, $b=-1$ and $c=3$
				Total 8 marks

(c) **ALT**	$ax^2+2bax+b^2a+c$		3	M1 for correctly expanding $a(x+b)^2+c$ to give $ax^2+2bax+b^2a+c$
	$2ba=-8$ and $b^2a+c=7$			M1 for a complete method (equating coefficients)
	Correct answer scores full marks (unless from obvious incorrect working)	$4(x-1)^2+3$		A1 allow $a=4$, $b=-1$ and $c=3$